

Geometry

Unit 6

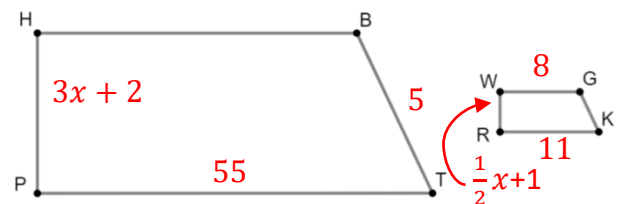
Lesson 13 Homework

Name Answer Key

Date _____

Period _____

Sheena, a party planner, is designing a banner for a birthday party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.



Sheena knows that $PT = 55$ inches (in), $HP = 3x + 2$ in, $WG = 8$ in, $WR = 0.5x + 1$ in, $BT = 5$ in, and $RK = 11$ in.

- What scale factor will map $HBTP$ to $WGKR$?

$$\frac{11}{55} = \frac{1}{5}$$

- What is the perimeter of $HBTP$, the main entrance banner?

$$\frac{3x+2}{\frac{1}{2}x+1} = \frac{5}{1}$$

$$3x + 2 = \frac{5}{2}x + 5$$

$$HP = 3(6) + 2$$

$$\frac{HB}{8} = \frac{5}{1}$$

$$3x + 2 = 5\left(\frac{1}{2}x + 1\right)$$

$$\frac{1}{2}x = 3$$

$$HP = 20$$

$$HB = 40$$

$$x = 6$$

$$P = 20 + 40 + 5 + 55 = 120 \text{ in}$$

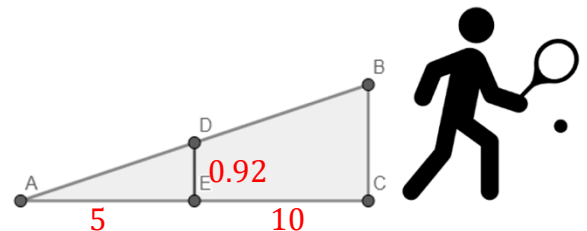
- What is the perimeter of $WGKR$, the photo booth banner?

$$\frac{120}{P} = \frac{5}{1}$$

$$5P = 120$$

$$P = 24 \text{ in}$$

Venus is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DE}$, and land 5 meters (m) away from the base of the net.



- Given that $\triangle ADE \sim \triangle ABC$, $AE = 5$ m, $DE = 0.92$ m, and $EC = 10$ m, what is the height, represented by $\overline{BC}$, of the tennis racket?

$$\frac{15}{5} = \frac{BC}{0.92}$$

$$BC = 0.92\left(\frac{15}{5}\right)$$

$$BC = 2.76 \text{ m}$$

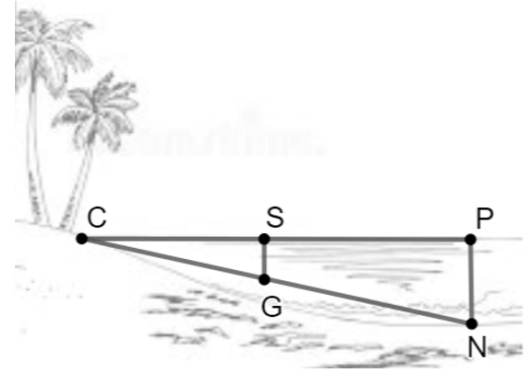
- What scale factor will map $\triangle ABC$ to $\triangle ADE$?

$$\frac{AE}{AC} = \frac{5}{15} = \frac{1}{3}$$

A beach front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SG}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 8$ feet (ft), $SG = (2x - 4)$ ft,

$PN = (5x + 8)$ ft, and $CP = 32$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?

$$\frac{CS}{CP} = \frac{8}{32} = \boxed{\frac{1}{4}}$$

7. What is the depth of the longer support post, represented by $\overline{PN}$?

$$\frac{5x+8}{2x-4} = 4$$

$$3x = 24$$

$$PN = 5(8) + 8$$

$$5x + 8 = 8x - 16$$

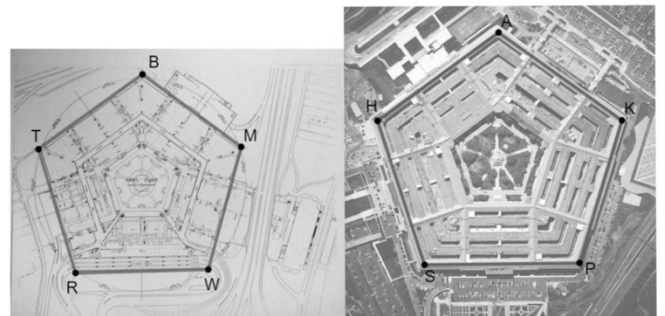
$$x = 8$$

$$\boxed{PN = 48 \text{ ft}}$$

8. What is the depth of the shorter support post, represented by $\overline{SG}$?

$$SG = 2(8) - 4 = \boxed{12 \text{ ft}}$$

Cesar wants to create a scale model of the Pentagon for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{5}$ of the scale model Pentagon, $AKPSH$. $TB = \left(\frac{2}{3}x + 4\right)$ centimeters (cm) and $HA = (6x + 4)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?

$$\frac{\frac{2}{3}x+4}{6x+4} = \frac{1}{5}$$

$$\frac{8}{3}x = 16$$

$$TB = \frac{2}{3}(6) + 4$$

$$P = 5 \cdot 8 \text{ cm}$$

$$\frac{10}{3}x + 20 = 6x + 4$$

$$x = 6$$

$$TB = 8 \text{ cm}$$

$$\boxed{P = 40 \text{ cm}}$$

10. What is the perimeter of the scale model Pentagon, $AKPSH$?

$$\frac{P}{40} = \frac{5}{1}$$

$$\boxed{P = 200 \text{ cm}}$$